Master's Thesis

Part 2/4 - Knowledge Base

RAG ingestion framework: Standardizing the heterogeneity of structured and unstructured enterprise data

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Abstract

The knowledge base outlines the theoretical and empirical foundations for a standardized RAG ingestion architecture annual financial reports to address the problem of enterprise data fragmentation. A systematic literature review followed by the PRISMA 2020 guidelines identified six academical sources underlying the primary challenges of heterogenous data fusion, advanced processing methods, LLM-based semantic mapping and the development of RAG frameworks. The synthesis claims that the preparation and matching of disparate data sources is critical. A standardized approach is necessary to reduce manual engineering effort and to increase reliability and contextual richness of enterprise data. Therefore, these findings are essential to define a comprehensive set of functional and non-functional requirements. The proposed artifact automates the transformation of disparate annual financial reports to provide contextual, traceable and reliable retrieved information by AI systems.

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1 Introduction

The thesis addresses the primary challenge to leveraging generative AI due to the lack of a standardized ingestion framework for heterogeneous enterprise data. Interviews showed that a significant amount of manual engineering work is necessary to map structured and unstructured data, such as annual financial reports, due to differences in semantics and syntax. This fragmentation creates a strategic liability known as “data debt”, which occurs when data is inactive or siloed and thereby limiting the scalability of intelligent systems. Consequently, these validated issues require the need for an artifact that standardizes data ingestion, provide semantically mapping and chunking to prepare data appropriately. This section synthesizes research to establish the theoretical foundations necessary to address these challenges.

1.1 Refined Problem Space

The problem space revealed that enterprises are facing challenges with a data landscape characterized by inactive and siloed heterogenous data, which prevents them from transforming fragmented information into contextual insights required for decision-making processes. Interviews confirmed that data heterogeneity is increasing and requires excessive manual engineering effort. Applying Kuckartz's (2019) qualitative content analysis, these issues were grouped into three properties.

- **Structural Heterogeneity:** Engineers must work through a landscape of both structured and unstructured data, which hinders efficient and consistent data processing.
- **Contextual Complexity:** A lack of common identifiers across different data sources, hinder systems to reliably match without intelligent contextual understanding.
- **Manual Engineering Effort:** Engineers spent a lot of time on manual analyses and individual coding logic, which results in enterprise data remaining inaccessible.

To address the identified problems, the artifact focuses specifically on annual financial reports and automates a standardized data ingestion pipeline, reducing the need for engineering effort. It uses semantic chunking and vectorization to resolve structural and contextual complexities.

1.2 Research Question

How can a standardized RAG ingestion architecture use LLM-based semantic mapping to automate the fusion of structured and unstructured annual financial reports, reducing manual engineering effort, while improving the reliability and contextual richness of the information retrieved?

2 Research Methodology

Since the design science research demands scientific rigor, the thesis conducts a systematic literature review in accordance with the PRISMA 2020 statement by Page et al. (2021). The methodology identifies the foundational theories and technical fundamentals necessary for a standardized RAG ingestion architecture.

2.1 Search Protocol and Strategy

A systematic search protocol was developed to capture the different perspectives that are relevant for the proposed artifact. The search strategy focused on the integration of AI-driven retrieval and heterogeneous data management within enterprises.

The following databases have been selected to conduct a structured search approach: FH Wien Library Catalogue, Google Scholar, Semantic Scholar and Research Rabbit, to ensure a comprehensive capture of peer-reviewed engineering and information systems literature. The timeframe was restricted to 2020–2025 to maintain latest technical progress with the rapid advancements in LLM capabilities, chunking and vectorization technologies.

The search was executed using structured Boolean strings to address the intersection of heterogenous data management and RAG frameworks:

1. **Heterogeneous Data/ Data Ingestion:** `("data ingestion" OR "data fusion") AND ("structured" AND "unstructured") AND ("Enterprise Data Fragmentation") OR ("Data Debt" AND "AI Ingestion") AND ("Heterogenous Data" OR "Data Heterogeneity")`
2. **RAG Framework:** `("RAG" OR "Retrieval Augmented Generation") AND ("Semantic Chunking" OR "Vectorization")`

2.2 Inclusion and Exclusion Criteria

To ensure research rigor, specific criteria were applied to filter the results of the systematic literature review. These criteria addressed the need to narrow the scope of the research and are summarized in the Table 1. The primary research focused on post-2020 peer-reviewed journals that deal with the processing heterogeneous data, data ingestion, data quality and RAG frameworks, which serve as the basis for the requirements of the artifact.

Category	Inclusion Criteria	Exclusion Criteria
Technical Scope	Focus on RAG architectures, enterprise LLM implementation, machine learning techniques, semantic chunking and vectorization methods.	General AI or Machine Learning papers that do not address data ingestion or RAG architectures
Data Ingestion	Focus on heterogenous data ingestion as structured, semi-structured, and unstructured data. Semantic mapping of heterogeneous data	Papers focusing exclusively on homogenous data sources as images, or simple SQL integration.
Data Quality	Focus on data quality challenges, data harmonization strategies, performance issues through chunking and vectorization	Theoretical papers on data quality that lack empirical applications or technical implementation details for enterprise environments.
Method	Peer-reviewed journals, conference papers or systematic analyses	Preprints, blog posts, white papers and online magazines
Timeframe	From 2020 to 2025	Prior to 2020
Language	Englisch and German	Any other language

Table 1: Inclusion and Exclusion Criteria for the Systematic Literature Review

2.3 Literature Selection Process

The systematic literature review according to the PRISMA 2020 diagram by Haddaway et al. (2022) was developed and is available in the Appendix, Figure 2.

The database searches yielded 146 records from scientific databases. After removing 34 duplicate entries, 112 records were screened and among those, 72 records were excluded for failing research criteria within the title or abstract. Of the 40 reports sought for retrieval, 4 could not be retrieved due to a paywall, leaving 36 reports to be assessed for full-text eligibility.

A total of 30 reports were excluded due to the following reasons:

- Not peer-reviewed, n=13
- Lack of sufficient technical depth regarding RAG architectures, n=7
- No focus on heterogeneous data, n=6
- Limited focus on LLMs, n=4

This review resulted in a final selection of 6 primary studies that have been included within the knowledge base. The selected studies provide the empirical validity and the basis for the foundational theories in the subsequent chapters.

3 Foundational Theories

This chapter establishes the theoretical foundations addressing the challenges of data heterogeneity by exploring strategies for data harmonization, semantic mapping and the fundamentals of RAG systems. The research identified principles as the scientific basis for the proposed artifact by emphasizing the critical value of advanced processing methods and data quality.

3.1 The Challenge of Data Heterogeneity

The rapid growth of big data, driven by technology, requires organizations to manage information from various sources, that includes structured, semi-structured, and unstructured data types. The heterogeneity of these data formats poses substantial challenges for decision-makers and demands the implementation of comprehensive data harmonization strategies to ensure consistent outcomes. Therefore, data harmonization is the critical field dedicated to unifying disparate data formats through structural alignment and advanced modern computational processing (Kumar et al., 2021).

According to Koukaras (2025) modern data ingestion demands hybrid Extract, Transform, Load (ETL) pipelines that incorporate flexible, AI-driven methodologies, such as natural language processing (NLP), to manage unstructured content, including text and images. Data integration is the unification of information from independent sources into a single, cohesive mapping schema. This process addresses the following three categories of data heterogeneity. At first, schema heterogeneity occurs when different datasets use different structures to represent the same information. This can include variations in attribute names or different data types, such as strings or integers for identification fields. Secondly, semantic heterogeneity occurs when data elements have similar names but different meanings. For example, the term "salary" could refer to gross annual income in one system and net monthly salary in another database. To mitigate these ambiguities, integration strategies must incorporate contextual mechanisms supported by metadata. The third instance-level heterogeneity is characterized by discrepancies in specific data values. Examples include conflicting date formats as YYYY-MM-DD versus DD-MM-YYYY, unit mismatches such as kilometers versus miles, and inconsistent identifiers. These discrepancies require rigorous, modern techniques. Consequently, integration platforms must dynamically handle diverse types of data heterogeneity, ensuring semantic integration and maintenance of data origin across both structured and semi-structured environments. The persistence of data silos, where information remains isolated, creates a significant barrier to achieve a holistic enterprise view. This fragmentation is often made worse by the absence of a common reference model, which makes it difficult for different systems to harmonize data effectively (Koukaras, 2025).

3.2 Strategies for Data Fusion and Semantic Mapping

The initial phase of data fusion usually includes schema matching, which identifies correspondences between elements of different schemas to establish an integrated framework. There are a variety of approaches for structured data matching, including name-based matching for linguistic similarities, structure-based matching for graph structures, and instance-based matchers for the analysis of data values and patterns. To harmonize data effectively, systems utilize a formal framework, technically known as an ontology, which explicitly defines the relationships and meanings of specific mapping concepts. This approach fosters a shared understanding that enables interoperability among disparate data sources by mapping their unique identifiers to a unified global schema definition (Koukaras, 2025).

Metadata management frameworks strictly complement these definitions, enhancing discoverability and governance by tracking data origin and enforcing standardization policies. Furthermore, managing unstructured data, especially textual datasets, requires advanced computational techniques that go beyond traditional storage capabilities. Organizations use harmonization approaches that incorporate natural language processing, machine learning, and deep learning to extract structured insights from raw text. These techniques are essential for transforming scattered, disparate information into uniform representations that support high-quality predictive analytics. The effectiveness of data harmonization frameworks is rigorously evaluated using specific performance metrics to measure accuracy based on precision, recall, and F-1 score across machine learning models. In addition to these measures, processing time is a critical factor in evaluating the efficiency of converting and fetching records from heterogeneous datasets. This is necessary to ensure that the chosen techniques are viable for enterprise applications (Kumar et al., 2021).

3.3 The anatomy of Retrieval Augmented Generation (RAG)

RAG is an architecture that leverages the intelligent potential of large language models with accurately retrieved content from an external database to generate more grounded and reliable output (Klesel & Wittmann, 2025).

According to Klesel & Wittmann (2025), the term RAG is defined as follows:

- **Retrieval:** The retriever identifies the most relevant information by calculating the similarity between the user query vectors and the document vectors.
- **Augmentation:** The system then generates an extended query from the retrieved context by using an augmentation template including the context in the final prompt.
- **Generation:** The generator produces the answer by processing the enriched query and the provided output may include links to the source of the documents.

The evolution of organizational information systems has led to the adoption of RAG as a primary method for overcoming the limitations of standard LLMs in enterprise environments. This architecture is particularly valuable because it mitigates the risk of hallucinations, where models produce plausible but false information, and addresses the issue by allowing access to external databases.

RAG systems increase transparency of AI interactions by providing clear source citations, which allows users to verify the retrieved information and ensures overall system trust. By bridging the gap between internal corporate knowledge and advanced linguistic capabilities, this framework provides a more efficient and reliable solution for complex enterprise search and decision-making tasks (Klesel & Wittmann, 2025).

Unlike traditional fine-tuning, which is often cost-prohibitive and time-intensive for large-scale models, RAG architectures offer a practical solution. It stores sensitive business information in vector databases, which can be used to provide real-time context for user queries. The effective deployment of an enterprise RAG system goes beyond the immediate retrieval process. It is governed by data homogenization, whereby the model extracts generalized knowledge from a consistent, unified dataset to solve diverse organizational problems. An orchestration layer such as “LangChain” can facilitate this integration by managing the complex interplay between the user prompt, the vector database, and the LLM (Jeong, 2023).

3.4 Advanced Data Processing Methods

Following the harmonization of heterogeneous data sources, it is essential to leverage semantic chunking and vector embedding techniques to convert the information into formats that can be processed by intelligent systems.

Semantic chunking is a sophisticated preprocessing strategy for RAG frameworks, which divides large documents into segments based on their conceptual meaning rather than on a fixed character count. Using natural language processing, this approach identifies logical breaks in the text, such as shifts in topic or meaning. It ensures that each retrieved segment keeps its actual context that is provided to the large language model and reduces the risk of generating inaccurate responses. Ultimately, semantic chunking is a process that enhances the precision of the retrieval phase by ensuring that the model processes cohesive units of information (Kshirsagar, 2024).

Vector embeddings transform data, such as text, images, or audio, into high-dimensional numerical values. This process maps information into a vector space where semantic relationships are represented by geometric proximities. One of the core properties of these embeddings is their fixed dimensionality, which ensures the resulting vector maintains a consistent

length for efficient processing. This mathematical foundation is critical for the retrieval phase of RAG systems because it enables the rapid identification of relevant context through similarity-based vector searches (Singh, 2025).

3.5 Data Quality Measures for RAG

Müller et al. (2025) claims that an effective implementation of RAG systems needs an approach to guarantee that data quality addresses four dimensions throughout the data ingestion pipeline. The intrinsic data quality is crucial to maintain accuracy and objectivity, to mitigate wrong information in the knowledge base. Contextual data quality ensures that the retrieved data is not only complete but also timely and strictly relevant to the specific use case to maintain high contextual reliability. The representational data quality is responsible for the consistency and clarity of data formats, ensuring that heterogeneous information is transformed into machine-readable structures that support coherent automated processing. Accessibility data quality addresses the technical challenges of accessing data, ensuring that information remains available and retrievable across systems. Semantic integration plays a pivotal role during the data transformation process, where the selection of an appropriate chunking strategy is critical to preserve the semantic integrity of the content. It is crucial that ingestion strategies are context-sensitive to mitigate content losses and prevent models from being unable to retrieve valuable key information (Müller et al., 2025).

4 Artifact Definition and Requirements

This chapter operationalizes the theoretical and empirical insights from the systematic literature review conducted in Chapter 3, thereby defining the artifact and its technical requirements using the MoSCoW method. The review emphasizes the critical importance of standardizing data preparation and pre-processing phases within the ingestion workflow, thereby establishing a reliable foundation for the effective deployment of RAG systems. Consequently, the functional and non-functional requirements derived from the previous chapters are consolidated in Table 2 to ensure alignment with the core properties of the research question.

4.1 Artifact Definition

The proposed artifact standardizes data processing and information retrieval for heterogeneous annual financial reports. By converting structured and unstructured data into machine-readable semantic context, the system maps disparate data labels that are aligned with the predefined global schema to maintain data consistency across data sources. The architecture uses semantic chunking to divide documents into contextual segments and employs data vectorization to transform information into vector embeddings.

This approach enables a RAG system to synthesize responses from heterogeneous data sources by effectively retrieving the requested information within the annual financial reports. The standardization of these core components ensures architectural scalability, significantly reducing manual engineering effort while increasing retrieval reliability and contextual richness.

4.2 Research Contribution

The primary contribution of this research is a standardized framework that resolves the heterogeneity between structured and unstructured data of annual financial reports to retrieve information across disparate data sources without manual engineering effort. In contrast to traditional RAG systems, which rely on high quality input data, this artifact introduces an automated semantic harmonization layer that utilizes LLM-based reasoning to align and map diverse data types. This innovation demonstrates that standardized data ingestion of heterogeneous sources is critical and a requirement for maintaining contextual reliability and traceability in complex enterprise environments to provide a fundamental base for decision-makers.

4.3 Functional Requirements

This section outlines the preliminary functional requirements (FR) derived from a synthesis of the systematic literature review and empirical insights collected through expert interviews.

FR1: Multi-Source Ingestion: The artifact must automatically load and extract content from heterogeneous sources, specifically structured and unstructured annual financial reports.

FR2: Semantic Flattening: The artifact must utilize an LLM to transform structured data into natural language to ensure they are semantically compatible with unstructured data.

FR3: Semantic Chunking: The artifact must break unstructured data as text into smaller units to preserve the semantic financial context.

FR4: Schema Mapping: The artifact must dynamically align labels to a predefined Global Schema using LLM-based semantic reasoning to ensure data consistency across sources.

FR5: Vectorization: The artifact must transform both the structured data and the segmented unstructured text into vectors and store them in a vector database for cross-source searching.

FR6: RAG Outcome: The artifact must execute a query across the vector storage to retrieve and generate a synthesized response that answers the user's question.

FR7: Orchestration Integration: The artifact should have an orchestration layer, such as LangChain, to manage interaction between the prompt, the vector database, and the LLM.

FR8: Metadata Enrichment: The artifact should have to automatically assign metadata to ingested data, including source trustworthiness flags and confidence levels.

4.4 Non-Functional Requirements

Based on the systematic literature review and empirical interview output, this section establishes the non-functional requirements (NFR) that define the quality attributes and performance of the proposed architecture

NFR1: Contextual Richness: The artifact must ensure that the context provided to the LLM is limited to the retrieved content to avoid hallucinations.

NFR2: Interpretability & Traceability: The artifact must have a grounded output and track every step of the ingestion and retrieval process, including the sources.

NFR3: Standardization: The artifact must have an ingestion logic that follows a repeatable procedure regardless of data formats, ensuring new data types can be integrated without changing the core pipeline.

Systematic Literature Review and Requirements Matching					
Properties	**Nr.**	**Requirements**	**Priority**	**References**	**Sources**
Reducing manual engineering effort	FR1	Multi-Source Ingestion	Must	"...requires organizations to manage information from various sources, that includes structured, semi-structured, and unstructured data types"	Kumar et al., 2021
	FR2	Semantic Flattening	Must	"schema heterogeneity occurs when different datasets use different structures to represent the same information…"	Koukaras, 2025
	FR4	Schema Mapping	Must	"...fragmentation is often made worse by the absence of a common reference model, which makes it difficult for different systems to harmonize data effectively."	Koukaras, 2025
	FR5	Vectorization	Must	which ensures the resulting vector maintains a consistent length for efficient processing.	Singh, 2025
	FR6	RAG Outcome	Must	"RAG...is governed by data homogenization, whereby the model extracts generalized knowledge from a consistent, unified dataset to solve diverse organizational problems."	Jeong, 2023
	FR7	Orchestration Integration	Should	"An orchestration layer such as “LangChain” can facilitate this integration by managing the complex interplay between the user prompt, the vector database, and the LLM"	Jeong, 2023
	NFR2	Interpretability & Traceability	Must	"RAG systems enhance transparency of AI interactions by providing clear source citations..and increases system trust."	Klesel & Wittmann, 2025
	NFR3	Standardization	Must	"The heterogeneity of these data formats poses substantial challenges for decision-makers and demands the implementation of comprehensive data harmonization strategies…"	Koukaras, 2025
Improving Reliability	FR3	Semantic Chunking	Must	"Semantic chunking is a process that enhances the precision of the retrieval phase by ensuring that the model processes cohesive units of information"	Kshirsagar, 2024
	FR6	RAG Outcome	Must	"This architecture is particularly valuable because it mitigates the risk of hallucinations,…"	Klesel & Wittmann, 2025
	NFR1	Contextual Richness	Must	"...integration strategies must incorporate contextual mechanisms supported by metadata."	Koukaras, 2025
	NFR2	Interpretability & Traceability	Must	"The intrinsic data quality is crucial to maintain accuracy and objectivity,.."	Müller et al., 2025
	NFR3	Standardization	Must	"The heterogeneity of these data formats poses substantial challenges for decision-makers and demands the implementation of comprehensive data harmonization strategies…"	Koukaras, 2025
Contextual Richness	FR1	Multi-Source Ingestion	Must	The persistence of data silos, where information remains isolated, creates a significant barrier to achieve a holistic enterprise view.	Koukaras, 2025
	FR3	Semantic Chunking	Must	"It ensures that each retrieved segment keeps its actual context that is provided to the large language model…"	Kshirsagar, 2024
	FR4	Schema Mapping	Must	"…,integration platforms must dynamically handle diverse types of data heterogeneity, ensuring semantic integration and maintenance of data origin across both structured and semi-structured environments"	Koukaras, 2025
	FR5	Vectorization	Must	"...is critical for the retrieval phase of RAG systems because it enables the rapid identification of relevant context through similarity-based vector searches	Singh, 2025
	FR8	Metadata Enrichment	Should	"To mitigate these ambiguities, integration strategies must incorporate contextual mechanisms supported by metadata. "	Koukaras, 2025
	NFR1	Contextual Richness	Must	"Contextual data quality ensures that the retrieved data is not only complete but also timely and strictly relevant to the specific use case to maintain high contextual reliability. "	Müller et al., 2025

Table 2: SLR Requirements Matching Table

5 Conclusion

The knowledge base has emphasized the foundational theories and methodological rigor required to develop a standardized RAG ingestion architecture. By synthesizing the literature review, the research provides a scientifically grounded response to the challenges that have been defined within the problem space. Therefore, every artifact requirement derived from the research and the empirical needs has been specified to ensure a scoped standardized RAG ingestion architecture within the next parts of the thesis.

The next part of the thesis focuses on the evaluation, development and conceptual plan of the proposed artifact. To validate the artifact, a test case demonstrates the automated fusion of heterogeneous annual financial reports, specifically structured JSON/XML files and unstructured PDF files. Triggered by a user request, the retrieval phase executes simultaneous vector searches to generate synthesized responses grounded in precise citations to ensure data traceability and reliability relative to the user's query.

The outcome of the artifact is presumably measured against three metrics derived from the research question. First, for the reduction in manual engineering effort, the system evaluates time to ingest data by comparing the time required for automated RAG processing with the hours required for manual engineering effort by programming Java components and conducting analyses. Second, to validate the reliability, the system uses a trust and hallucination rate assessment, which quantifies the extent of the generated insights that are grounded in the retrieved source data. Third, to evaluate contextual richness, the artifact measures the semantic mapping accuracy through precision and recall of metadata alignment. This metric quantifies the number of structured and unstructured data field correlations achieved without human intervention and demonstrates the system's ability to preserve complex relationships within heterogeneous data.

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List of Tools Used

AI tool	Purpose	Concerned parts of the work	Notes
Deepl Translator https://www.deepl.com/de/translator	Translation of text passages	Whole document	
Deepl Writer https://www.deepl.com/de/write	Rephrase and grammar check	Whole document	
Gemini http://gemini.google.com/	Literature research	Foundational Theories, Systematic Literature Review	Researching literature accessible via the used databases; Early stage SLR explanation and brainstorming phase
Gemini http://gemini.google.com/	Interview content analysis	Interviews	Support with the Kuckartz (2019) content analysis of the interview transcripts

Glossary

Term	Abbreviation	Definition
Artificial Intelligence	AI	Technology that enables computers and machines to simulate human learning (Source: IBM: ”What is AI?)
Application Programming Interface	API	Set of rules and protocols that enables software applications to communicate (Source: IBM: “What is an API”)
Extract Transform Load	ETL	A data integration process that combines, cleans and organizes data from multiple sources into a single, consistent data set for storage in a data warehouse (Source: IBM: What is ETL?)
Large Language Model	LLM	Machine Learning Models that comprehend and generate human language (Source: Cloudfare: “Was ist ein großes Sprachmodel?”)
LangChain	LangChain	LangChain ist ein Open-Source-Orchestrierungs-Framework für die Entwicklung von Anwendungen mit großen LLMs (Source: IBM: Was ist LangChain)
Natural Language Processing	NLP	A subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language (Source: IBM: What is NLP?)
Retrieval Augmented Generation	RAG	Architecture for optimizing a model by connecting it with external knowledge bases to deliver a higher quality (Source: IBM: “What is RAG?”)

Appendix A: PRISMA 2020 Flow Diagram

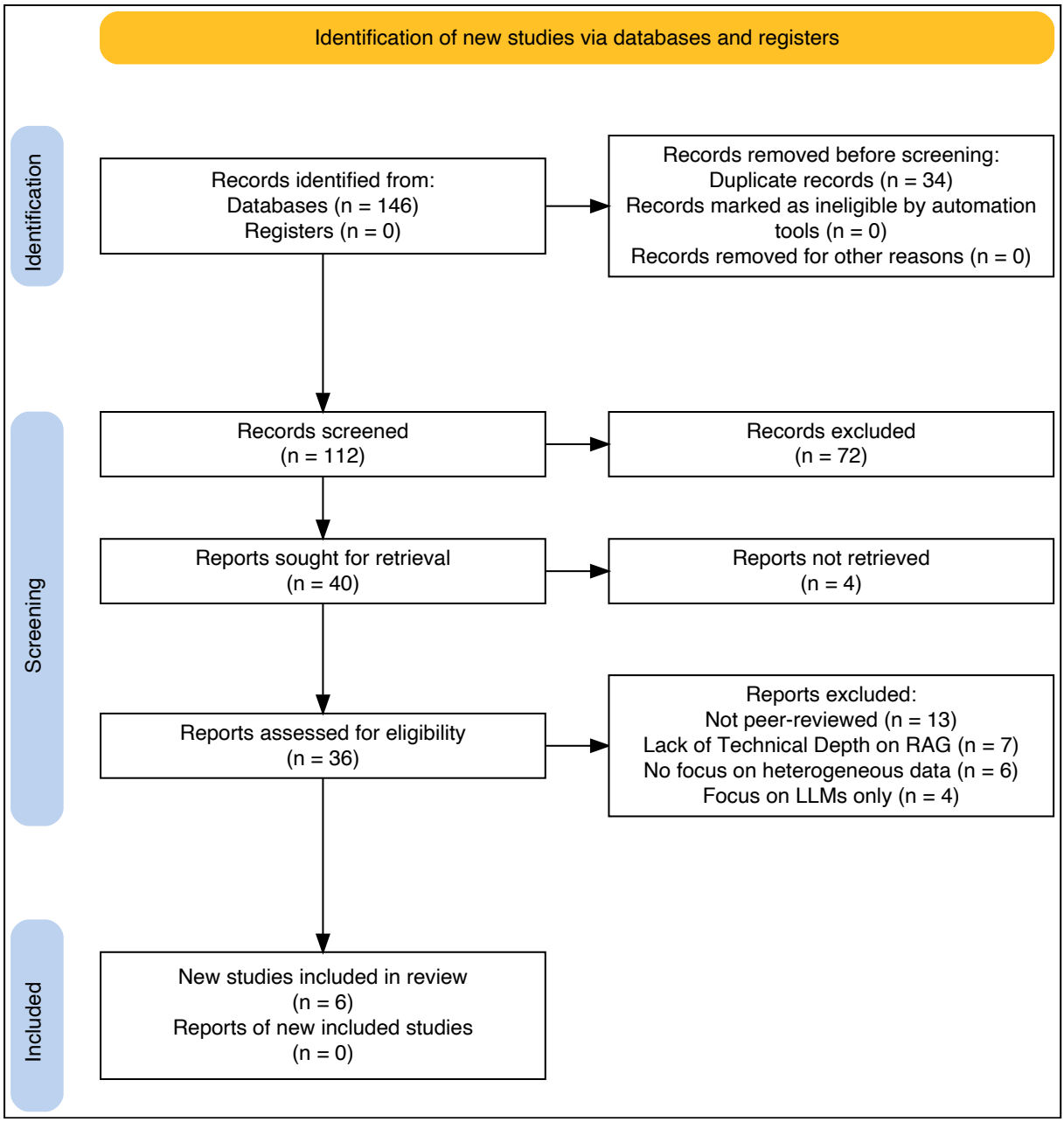

Figure 1:PRISMA 2020 diagram for the systematic literature review